Assignment 2

Technical Analysis using R, Preliminary Stage (5%)

DUE DATE

Session 3

OBJECTIVE

- Set up a GitHub account and repository.
- Install R and RStudio on a local PC.
- Download and install R packages.
- Write utility functions to read and compute statistics on data.
- Configure and initialize the technical analysis process.
- Display imported data and calculated statistics.
- Document the setup and analysis process.

DESCRIPTION

In this assignment, you will set up a GitHub account and repository to store all of your assignments in this course. This assignment is part of a three-stage project that will span over 3 assignments in this course. In this stage, will install R and RStudio, create a text file with stock symbols, download necessary R packages for technical analysis, and write functions to import and compute basic statistics on the stock data. Finally, you will display the imported data along with calculated statistics. You will document every step of your setup process and include screenshots of each step in your assignment report document. In your repository, you are expected to create one subdirectory for each assignment and place all your files in their respective folders.

INSTRUCTIONS

1. **Setting up a GitHub Account and Repository:**
    - If you don't have a GitHub account, sign up for one.
    - Follow the instruction on the GitHub site.
      For installation guide go to https://github.com/git-guides/install-git.

For setup go to https://docs.github.com/en/get-started/quickstart/set-up-git.

- Alternatively, you may want to watch a video to get started.
  For a quick video, watch “GitHub Tutorial - Beginner's Training Guide” by Anson Alexander at https://youtu.be/iv8rSLsi1xo.
- For a longer crash course, watch “Git and GitHub for Beginners - Crash Course” by freeCodeCamp.org at https://youtu.be/RGOj5yH7evk.
- Create a new repository named 'TechnicalAnalysis' to store your assignments.
- Document the entire setup process by taking screenshots and saving them in your assignment report.

2. **Installing and Configuring R and RStudio:**
   - Download and install R and RStudio on your computer's operating system.
   - Follow the instruction on RStudio website at https://cran.rstudio.com/ to download and install R.
   - Alternatively, watch the following videos to assist you with the installation steps.
     Watch “How to install R and install R Studio” by R Programming 101 at:
     https://www.youtube.com/watch?v=orjLGFmx6l4
     or Watch “How to download and install R and RStudio” by Dynamic Data Script at https://youtu.be/TFGYlKvQEQ4.
   - Capture screenshots of each installation step and include them in your report.
3. **Downloading and Installing R Packages for Technical Analysis:**
   - Install necessary R packages like 'quantmod,' 'TTR' (Technical Trading Rules), and any other relevant packages using the 'install.packages()' function.
   - Follow the instruction on CRan website at https://cran.rstudio.com/doc/manuals/r-release/R-admin.html#Installing-packages for a guide on how to install packages in R.
   - Alternatively, watch “How to import data and install packages” by R Programming 101 at https://youtu.be/e8B9YU_M5FM.
   - Ensure you document the package installation process by taking screenshots for each step.
4. **Project Configuration and Initialization:**
   - Create a text file named 'portfolio.txt.'
   - List the stock symbols, one per line, in the 'portfolio.txt' file to represent your individual portfolio.
5. **Writing a Function to Import and Load Stocks Data:**

- Develop a function, e.g., 'load_stock_data()', which reads the 'portfolio.txt' file and imports the stock data for each symbol using the 'quantmod' package.
- For a tutorial to work with quantmod package, you may watch “QuantMod in R Part 1” by Ed boone at https://www.youtube.com/watch?v=AXoWjVAlnzY.
- Make sure to load the data into R and store it as separate data frames for each stock symbol.

6. **Writing a Functions to Compute Basic Statistics:**
    - Create a function, e.g., 'calculate_statistics()', that takes each stock's data frame as input and calculates the required statistics (moving average, mean, mode, median, and standard deviation) using functions from the base R or other relevant packages.
    - Store the calculated statistics for each stock in a structured format, such as a data frame or a list.
7. **Write Utilities to Display Output:**
    - Write your customized functions to display the loaded data frames for each stock, showing their respective information.
    - You are encouraged to find out how many different ways the stocks data may be displayed. Use a web search to find out how stock data is presented. You may wish to browse through Yahoo finance page at https://finance.yahoo.com/ to familiarize yourself with how the stocks market data are presented.
8. **Documentation:**
    - Ensure you follow the step-by-step instructions to complete the assignment successfully.
    - Remember to include all necessary screenshots and documentation in your report for a comprehensive submission.

SUBMISSION INSTRUCTIONS

Deliverables:

1. GitHub repository with the name 'TechnicalAnalysis.'
2. Report document with detailed instructions, including screenshots, for each step.
3. The 'portfolio.txt' file containing the list of stock symbols.
4. R script or R Markdown file with the functions for loading data and computing statistics.

5. R script or R Markdown file with code to display the imported data and visualizations.

Submission Instructions:

1. Report document: Submit the assignment report in and MS-Word containing the screenshots of each step. Place (upload) the reports within your repository in their respective folders.
2. R script/R Markdown files: Submit the R script or R Markdown files containing the required functions and data display code. Place the codes within your repository in their respective folders.
3. GitHub repository link: Share the GitHub repository link where the assignment files are stored. To submit the GitHub link, create a cover page for your assignment in MS-Word with your details. Write the title of the assignments and place the link to your repository at the end. You should following the naming convention of your college in the format given below.

Work must be submitted in the correct file type and be properly labelled as per the College naming convention:

NAME_COURSE_ASSIGNMENT. E.g. XuXiaLing_FM50D_A01.

GRADING CRITERIA

Assignment Value: **5%**

Grading Criteria	Grade
GitHub Account and Repository Setup Creating a GitHub account and repository, and documenting the setup process with screenshots.	/10
RStudio Installation and Setup Downloading and installing R and RStudio with step-by-step screenshots.	/15
R Packages Installation Installing necessary R packages for technical analysis and providing documentation.	/15
Project Configuration and Initialization Creating a text file named 'portfolio.txt' and listing stock symbols in it.	/5

Function for Importing Stock Data Implementing a function to read the 'portfolio.txt' file and import stock data for each symbol using the 'quantmod' package.	/15
Function for Computing Statistics Developing functions to calculate basic statistics (moving average, mean, mode, median, and standard deviation) for each stock's data.	/20
Data Display Output Displaying the loaded data frames and creating visualizations of the data and statistics.	/10
Documentation Following instructions, providing necessary screenshots, and comprehensive documentation.	/10
Total	**/100**

RUBRIC

Grading Criteria	Insufficient	Acceptable	Good	Excellent
GitHub Account and Repository Setup	Setup process not documented or missing	Setup process partially documented	Setup process documented with few errors	Setup process fully documented without errors
RStudio Installation and Setup	Installation steps incomplete or incorrect	Complete installation with some errors	Complete installation with minor issues	Accurate and complete installation with no errors
R Packages Installation	Missing package installations	All required packages installed	All required packages installed with some documentation	All required packages installed with clear documentation
Project Configuration and Initialization	File not created or stock symbols missing	File created with some symbols listed	File created with all symbols listed	Well-organized file with clear formatting
Function for Importing Stock Data	Function not created or not reading data correctly	Function created with errors	Function reads data correctly for some symbols	Function reads data correctly for all symbols
Function for Computing Statistics	Function not created or incorrect calculations	Function created with some errors	Function calculates some statistics correctly	Function calculates all statistics correctly
Data Display Output	Data display missing or incorrect	Partial data display with errors	Data displayed for some stocks with minor errors	Data displayed correctly for all stocks
Documentation	Poorly documented or missing screenshots	Adequate documentation with few screenshots	Well-documented with some minor omissions	Highly detailed documentation with screenshots
Additional Comments	The total score of the assignment is out of 100 points. The instructor can use this rubric to assess the students' work and assign appropriate grades based on their performance in each criterion.			